

INVESTIGATION 8.1

Ticker timer tapes

AIM To record and analyse motion using a ticker timer

Materials:

ticker timer
power supply
scissors
G-clamp
ticker tape (in 60 cm lengths)

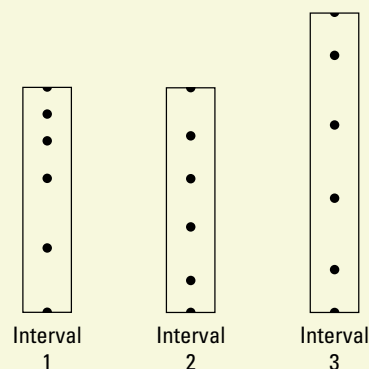
METHOD AND RESULTS

- ▶ Clamp the ticker timer firmly to the edge of a table or bench so that you will be able to pull 60 cm of ticker tape through it. Connect the ticker timer to the AC terminals of the power supply and set the voltage as instructed by your teacher.
- ▶ Thread one end of the ticker tape through the ticker timer so that it goes under the carbon paper disc.
- ▶ Turn on the power supply and check that the ticker timer leaves a black mark on the ticker tape.
- ▶ Hold the end of the ticker tape and walk away from the ticker timer so that the ticker tape moves through at a steady speed.
- ▶ Remove the ticker tape and mark off the first clear dot made and every fifth dot after the first. (There should be four dots between each of the marked-off dots on the ticker tape.)

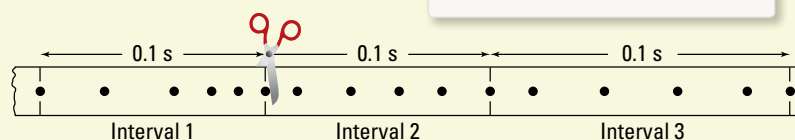
- 1 Measure the distance travelled during each 0.1 s interval and write it on your tape. Label the intervals as interval 1, interval 2, interval 3, etc.
- ▶ Cut your ticker tape into 0.1 s intervals and glue the strips in order onto a sheet of paper. Each strip shows the distance travelled during a 0.1 s time interval. The graph therefore shows how the speed changed with time.

DISCUSS AND EXPLAIN

- 2 How much time elapsed between the printing of the first clear dot and the last dot marked off?
- 3 Calculate the average speed for the motion that took place between the printing of the first clear dot and the last marked dot.
- 4 Calculate the average speed during each 0.1 s interval.
- 5 Did you succeed in keeping your speed steady?



Using ticker tape to plot a graph



UNDERSTANDING AND INQUIRING

REMEMBER

- 1 Explain the difference between instantaneous speed and average speed. Use an example to support your explanation.
- 2 Which methods are used by the police to measure:
(a) average speed
(b) instantaneous speed?

THINK

- 3 Make a list of reasons why a speedometer reading might not be accurate. Include anything that could change the diameter of the vehicle's tyres.
- 4 After being phased out, digitectors are making a comeback in police detection of speeding drivers.

Suggest advantages they might have over radar and laser guns.

PROCESS AND ANALYSE

- 5 Calculate the average speed during the second and third 0.1 s intervals of the ticker tape shown in Investigation 8.1.

INVESTIGATE

- 6 Use data-logging equipment with a motion detector or light gates to record the motion of a toy car or cart down a slope. Use the software to produce a graph of distance versus time and a graph of speed versus time. Comment on the shape of your graphs.

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8.1 Speed and velocity
8.2 Ticker tapes